

Multi-dimensional numerical analysis of flow instabilities in 3D-shaped honeycomb absorbers

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ABSTRACT

Considered a valid alternative to other receiver types, open volumetric receivers for central tower power plants have been developed and researched for multiple decades. 3D-Shaped absorber geometries, made possible by modern manufacturing technologies, particularly offer the promise of high thermal efficiencies. Yet, volumetric absorbers carry the inherent risk of flow instabilities, which can lead to a total failure of the receiver due to local overheating. Therefore, a numerical, multi-dimensional methodology for the flow stability analysis of volumetric absorbers is proposed. The first stage of the methodology applies an 1D LTNE continuum model to quantify the general pressure loss characteristic of an absorber and identifies critical pressure drop levels with potential for flow instability. These levels are tested as part of the second analysis stage using a discrete 3D CFD model against different irradiation and flow perturbations. If the absorber is able to compensate the disturbances, its flow behavior is deemed stable. Otherwise, the absorber geometry is considered susceptible to local overheating due to flow instability.

The new methodology has been validated against two reference absorbers with known stable resp. unstable flow behavior. Further, the new approach was applied to analyze the flow stability of two 3D-shaped honeycomb absorber geometries. It was shown that the geometric features of 3D-shaped absorbers can directly improve the flow stability, but do not prevent instabilities automatically. Therefore, a flow stability analysis should be included in design/optimization processes of new volumetric absorbers alongside a maximization of the thermal efficiency.

1. Introduction

Open volumetric receivers (OVR) have been developed for more than 25 years in parallel to the prevailing molten salt technology as a realistic alternative for central receiver power plants on an industrial scale. Most work in the past and recently has been dedicated to finding suitable porous structures as volumetric absorbers. However, volumetric absorbers carry the risk of flow instabilities, where self-reinforcing local overheating can lead to total failure of the structure. Receiver failures caused by local overheating have been observed during experimental tests of different volumetric absorbers in the past (e.g. Hoffschmidt, 1997; Buck, 2000; Hoffschmidt et al., 2001). Kribus et al. (1996) were the first to show, using a theoretical one-dimensional model, the fundamental tendency towards flow instabilities for those volumetric absorbers that exhibit a linear dependence of the pressure loss on the flow velocity. Hoffschmidt (1997) and Pitz-Paál et al. (1997) developed a quasi-three-dimensional, which allowed to model

the effects of radial heat transfer and non-homogeneous irradiation and included linear and quadratic pressure loss modeling based on the Darcy–Forchheimer equation. After the investigation of different types of volumetric absorbers, they concluded that absorbers with a dominantly quadratic pressure loss exhibit stable flow behavior and radial heat transfer reduces the probability of flow instabilities. Buck (2000) focussed on flow instability in solar–chemical receiver–reactors. They derived that solar–chemical reactors are generally more susceptible for flow instabilities than volumetric receivers, but different design measures aimed at modifying the pressure drop characteristic of the reactor can increase the flow stability. Becker et al. (2006) utilized a theoretical and a numerical model to study flow stability in different porous materials, and identified materials with a high thermal conductivity and a predominantly quadratic pressures loss behavior as the preferable materials for solar applications. Singh and Chandra (2018) analyzed the pressure loss characteristic of an OVR based on circular cylinders which showed no tendency for flow instabilities.

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¹ Within the context of this publication, honeycomb absorber refers to absorber structures characterized by parallel channels (opposed to e.g. porous foams).

Nomenclature

Acronyms

CFD	Computational fluid dynamics
HiTRec	High Temperature Receiver
LTNE	Local thermal non-equilibrium
OVR	Open volumetric receiver
SiSiC	Siliconized silicon carbide

Greek Symbols

α	Absorptivity (–)
Δp	Pressure loss (Pa)
ϵ	Emissivity (–)
κ	Heat conductivity ($\frac{W}{mK}$)
μ	Dynamic viscosity (Pa s)
ρ	Density ($\frac{kg}{m^3}$)
σ	(Linear) stress tensor ($\frac{N}{m^2}$)
σ_{sb}	Stefan–Boltzmann constant ($\frac{W}{m^2 K^4}$)
Ψ	Viewfactor (–)

Roman Symbols

G	Irradiance ($\frac{W}{m^2}$)
h	Specific enthalpy ($\frac{J}{kg}$)
I	Unit tensor (–)
I	Incident irradiance ($\frac{W}{m^2}$)
J	Surface radiosity ($\frac{W}{m^2}$)
K	Kinetic energy ($\frac{J}{kg}$)
L	Absorber length (m)
N	Number of absorber unit cells (–)
p	Pressure (Pa)
p^*	Normalized pressure loss (–)
$\dot{q}$	Heat flux ($\frac{W}{m^2}$)
T	Temperature (K)
T^*	Normalized outlet temperature (–)
u	Velocity vector ($\frac{m}{s}$)
W	Width (m)

Subscripts

abs	Absorber
air	Air
ap	Aperture
cond	Conduction
f	Fluid
in	Inlet
k	Radiation index
lvl	Level
max	Maximum
out	Outlet
r	Radiative
s	Solid
sec	Section
sol	Solar
surf	Surface
tot	Total

Aiming at improving the thermal efficiency of open volumetric receivers, recent research focuses on 3D-shaped volumetric absorbers, but these absorbers have yet to be investigated with regards to flow

stability. However, a flow stability analysis should be part of the design process of new 3D-shaped absorbers, because improved thermal efficiencies generally lead to more unstable flow behavior as shown by Hoffschmidt (1997). 3D-Shaped absorbers are volumetric structures where the geometry purposefully varies along the main flow direction. As a result, important absorber characteristics (e.g. the porosity) also change throughout the absorber, potentially improving the thermal efficiency. Investigated 3D-shaped absorbers include layered ceramic foams (Roldán et al., 2014; Chen et al., 2015; Zaversky et al., 2018), metal wire meshes (Ávila-Marín et al., 2018), stacked metal sheet absorbers (Pabst et al., 2017; Broeske et al., 2022), and honeycomb absorbers¹ (Gomez-Garcia et al., 2015; Alberti et al., 2016; Cagnoli et al., 2016; Capuano et al., 2017; Broeske et al., 2022). The listed studies represent only an excerpt of the research in the field of open volumetric receivers, more information on the development of the OVR technology can be found in the reviews by Ávila-Marín (2011) and Lipiński et al. (2021).

A new two-stage methodology is introduced within this publication specifically designed for the investigation of 3D-shaped volumetric absorbers' flow stability. For the first time, a discrete three-dimensional CFD model is used to analyze flow instability as part of the new methodology, which includes lateral radiative and conductive heat transfer between absorber unit elements. By applying the methodology to 3D-shaped honeycomb absorbers, it is demonstrated how volumetric absorbers can be directly designed to improve their flow stability.

2. Methodology

The new approach to investigate flow instability in volumetric absorbers is divided into two stages. First, a fast, one-dimensional continuum model is used to analyze the general pressure loss characteristic of an absorber structure. From the pressure drop characteristic, pressure drop levels with potential for flow instability are identified, which are further examined in the second analysis stage. By employing a three-dimensional CFD model, the flow behavior of an absorber geometry is tested against irradiation and flow perturbations in order to determine if the volumetric absorber operates stably at the selected levels.

2.1. First stage - 1D continuum model

Volume-averaged methods, also called continuum models, are common approaches for the simulation of volumetric absorber geometries. Opposed to unit cell models (cf. Section 2.2), the absorber geometry is not directly modeled by the continuum approach. Rather, the heat transfer and flow mechanisms are governed by equations relying on effective properties. These properties (e.g. convective heat transfer coefficients) are determined experimentally or calculated from auxiliary numerical simulations. The review by Ávila-Marín et al. (2019) gives a thorough summary of the different continuum models used for the simulation of volumetric absorbers. With an one-dimensional, local thermal non-equilibrium (LTNE) continuum model previously developed by the authors (Broeske et al., 2022), the general pressure loss characteristic of the investigated absorbers is evaluated. Detailed information about the 1D model, e.g. on the governing equations, the boundary conditions and the radiation modeling, can be found in the previous publication by the authors (Broeske et al., 2022).

Based on the studies by Pitz-Paal et al. (1997) and Becker et al. (2006), flow instabilities can occur if a single pressure drop level is associated with multiple air outlet temperature values. Such a pressure drop level would enable the absorber structure to reach different temperature levels locally compared to the overall operating point of the OVR. If one of the associated temperature values is above the maximum allowable temperature of the absorber material, structural failure of the absorber due to local overheating becomes possible.

The pressure drop characteristic of each investigated absorber is derived by running simulations with the continuum model for a wide

range of different massflow density values. For the evaluation, a normalized air outlet temperature T^* and a normalized pressure loss p^* are introduced as defined by Pitz-Paal et al. (1997):

$$T^* = \frac{T_{f,out} - T_{in}}{T_{in}} \quad (1)$$

$$p^* = \frac{p_{in}^2 - p_{out}^2}{p_{in}} \frac{\mu_{in}}{\rho_{in} L I_{ap}} \times 10^8 \quad (2)$$

The normalization is based on the incident solar irradiation I_{ap} , the absorber depth L , and the inlet pressure p_{in} and temperature T_{in} . The normalized pressure loss is expected to fall monotonically with an increasing normalized outlet temperature for stable volumetric absorbers. Accordingly, the characteristics of absorbers with potential for flow instability would show regions with non-monotonic behavior. Within these regions, pressure drop levels associated with multiple outlet temperatures can be identified. These pressure drop levels are then selected for the second stage of the analysis.

2.2. Second stage - 3D CFD model

The continuum model used for the first analysis stage is based on the central assumption of a laterally uniform flux and temperature distribution allowing the reduction to one dimension. However, flow instabilities in volumetric absorbers are specifically associated with non-uniform flux and flow conditions (cf. Pitz-Paal et al., 1997; Becker et al., 2006). A three-dimensional CFD model is therefore adopted as the second stage of the instability analysis. Similar to volume-averaged approaches, 3D CFD models are often used for the calculation of the thermal efficiency of absorber geometries (cf. Ávila-Marín et al., 2019). In order to limit the needed computation time, uniform behavior is generally assumed again for the efficiency calculation and the absorber geometries are reduced to so called “unit cells”, the smallest representative elements (e.g. a single channel for honeycomb absorbers). This assumption is dropped for the flow stability analysis and a horizontal row of N_{tot} unit cells is simulated instead of a single absorber unit cell. The details of the 3D CFD model are shown in Appendix.

For each pressure drop level selected during the first stage, a series of simulations with the stationary 3D model are conducted. Through the introduction of disturbances, the flow behavior of the absorber geometry is tested w.r.t. its stability similar to the numeric model by Becker et al. (2006). First, a reference case with an uniform incident solar irradiation and an uniform pressure drop for all absorber channels is calculated. Constant, uniform temperature distributions are used as the initial values. For the introduction of the perturbations, the row of absorber unit cells is divided into three horizontal sections (similar to Pitz-Paal et al., 1997), each containing N_{sec} unit cells. Both, irradiation and flow perturbations are tested mimicking non-homogeneous conditions caused by e.g. cloud-passages or a temporary blockage of absorber channels. Starting again from uniform temperature distributions, a solution with an active irradiation disturbance is created by increasing the incident solar irradiation for the left section, reducing the irradiation for the right section, while using the base flux level for the middle section. This solution is then applied as the starting value for another simulation, where the incident irradiation is reverted to the homogeneous distribution of the base case for all sections. Similarly, a solution with an active flow disturbance is created starting from uniform temperature distributions. Instead of setting a uniform pressure drop for the absorber channels, massflow density levels are enforced for each section. For the middle and right section, the massflow density is set to the value of the base case solution. The left section is assigned a significantly lower massflow density simulating a (partial) blockage. The resulting solution is used as the starting value for another simulation, where the flow disturbance is reverted by prescribing the uniform pressure drop of the base case again. For the final two simulations, simultaneous irradiation and flow perturbations are considered. An overview of the simulation cases is given in Table 1,

Table 1

Overview of the different 3D simulation cases.

Case	Start values	Disturbed irradiation	Disturbed flow
Base	constant values	No	No
Ir-A	constant values	Yes	No
Ir-D	Ir-A	No	No
Fl-A	constant values	No	Yes
Fl-D	Fl-A	No	No
Ir+Fl-A	constant values	Yes	Yes
Ir+Fl-D	Ir+Fl-A	No	No

a suffix “A” indicates a simulation with active perturbation and a suffix “D” a simulation with deactivated perturbation.

Based on the results by Becker et al. (2006), the irradiation and flow disturbances should not have any effect on the calculated solutions for stable absorber geometries. The simulations with deactivated perturbations are therefore expected to converge to the solution of the base case for stable absorbers. The results for volumetric absorbers with potential for flow instabilities are anticipated to vary between the base case and the perturbed cases.

3. Results

3.1. Reference absorbers

In order to validate the new methodology, two monolithic honeycomb absorbers have been selected for the first simulation series. The HiTRec absorber is manufactured out of siliconized silicon carbide (SiSiC) and consists of simple, square channels with a channel width of 2.0 mm and a wall strength of 0.8 mm (Hoffschmidt, 1997). The HiTRec absorber has been continuously proven to be a robust volumetric absorber without any known flow stability issues throughout experiments (e.g. Hoffschmidt et al., 2003) and the operation of the Solar Tower Jülich demonstration power plant (Koll et al., 2009). The second design, the CorRec absorber, has exhibited stability issues during experimental testing, which even lead to local melting of the absorber structure (Hoffschmidt et al., 2001; Fend et al., 2004). Similar to the HiTRec design, the CorRec absorber is also composed of square channels, but has a channel width of 1.12 mm and a wall strength of 0.13 mm and is made out of the ceramic material Cordierit (Hoffschmidt, 1997). The finer channels and thus higher porosity of CorRec design indicate a more linear pressure drop behavior compared to the less porous HiTRec absorber, which in combination with the significantly lower heat conductivity of Cordierit compared to SiSiC can explain the tendency towards flow instability (cf. Pitz-Paal et al., 1997; Becker et al., 2006). The aim of the first simulation series is to show that the new methodology can correctly depict the stable resp. unstable flow behavior of the two honeycomb absorbers.

Aperture² (normal) irradiance levels of $I_{ap} = 0.5 \frac{\text{MW}}{\text{m}^2}$, $1.0 \frac{\text{MW}}{\text{m}^2}$ and $1.5 \frac{\text{MW}}{\text{m}^2}$ are considered, where the average angle of the incident radiation is parallel to the main axis of the absorber channels and the maximum opening angle is 15°. The ambient pressure p_{in} and temperature T_{in} are set to 101 325 Pa and 100 °C. As the first stage of the flow stability analysis, simulations with varying massflow densities were carried out using the 1D continuum model for each flux level and both geometries. The resulting normalized pressure loss p^* (Eq. (2)) values are plotted over the normalized outlet temperatures T^* (Eq. (1)) in Fig. 1.

As seen in the zoomed, right view in Fig. 1, the CorRec absorber exhibits areas with a non-monotonous pressure loss behavior for all

² Within the context of this publication, the aperture is defined as the plane containing the absorber front.

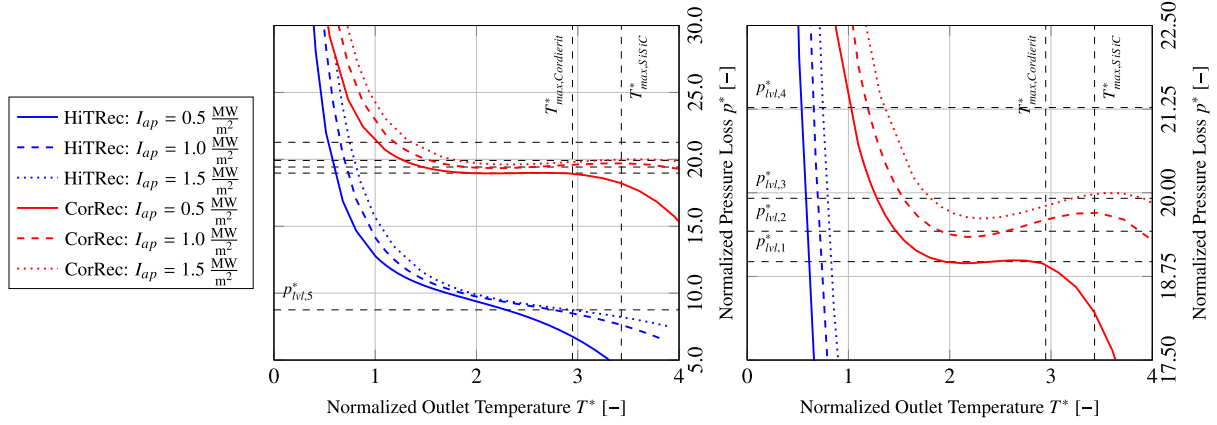


Fig. 1. Stage 1 results — normalized pressure loss over normalized outlet temperature for the HiTRec and CorRec absorbers (left plot: full pressure loss range; right plot: zoomed view of the critical zones for the CorRec absorber).

Table 2

Pressure loss levels selected for the second stage of the flow stability analysis.

Level	Absorber	I_{ap} [$\frac{\text{MW}}{\text{m}^2}$]	p_{lvt}^* [–]	Δp_{lvt} [Pa]
1	CorRec	0.5	19.0	82.0
2		1.0	19.4	168.0
3		1.5	21.3	257.5
4		1.0	19.8	184.0
5	HiTRec	1.5	8.75	113.5

three flux density levels. Within these regions, three different outlet temperature levels correspond to the same pressure loss value, a clear indicator for unstable flow behavior. For the two higher flux levels, the highest temperatures of the unstable regions extend beyond the maximum temperature of 1200 °C for Cordierit (Hoffschmidt, 1997), which is indicated by the left dashed line in Fig. 1, demonstrating the danger for overheating. Oppositely, the results for the HiTRec geometry show a monotonously decreasing pressure loss with increasing outlet temperatures. Therefore, the model predicts no instability problems for the HiTRec absorber.

Five pressure drop levels (Table 2) have been selected for the second stage of the flow stability analysis as indicated by the horizontal dashed lines in Fig. 1. One pressure drop value was chosen for each of the instability regions for the CorRec absorber. Additionally, a fourth pressure drop level was selected for $I_{ap} = 1.0 \frac{\text{MW}}{\text{m}^2}$, which is expected to correspond to stable operating point for the CorRec absorber similar to the pressure value picked for the HiTRec design.

For each pressure drop level, seven 3D simulations as defined in Table 1 have been carried out. The irradiation and flow perturbations were implemented as described in Section 2.2 and a horizontal row of twenty four absorber channels was considered for both geometries (cf. Section 3.2). The 3D simulation results are presented in Fig. 2. Each row corresponds to one simulation, showing normalized outlet temperatures based on massflow-weighted averages for the left, mid, and right third, and for all 24 channels.

All five base cases show homogeneous outlet temperatures as is expected for the uniform irradiation and pressure boundary conditions. For cases with active disturbances (“Ir-A”, “Fl-A”, “Ir+Fl-A”), the normalized temperatures vary for the three sections, showing the effect of the applied perturbations. The behavior of the cases with the deactivated disturbances (“Ir-D”, “Fl-D”, “Ir+Fl-D”) however differs between the HiTRec and CorRec absorber. For the HiTRec design, the three cases with deactivated disturbances yield a homogeneous outlet temperature distribution again, which matches the results of the base case. This corresponds to the behavior expected for stable volumetric absorbers. The CorRec design only shows a stable behavior for the pressure drop of $\Delta p = 184.0$ Pa and $I_{ap} = 1.0 \frac{\text{MW}}{\text{m}^2}$. The results for the

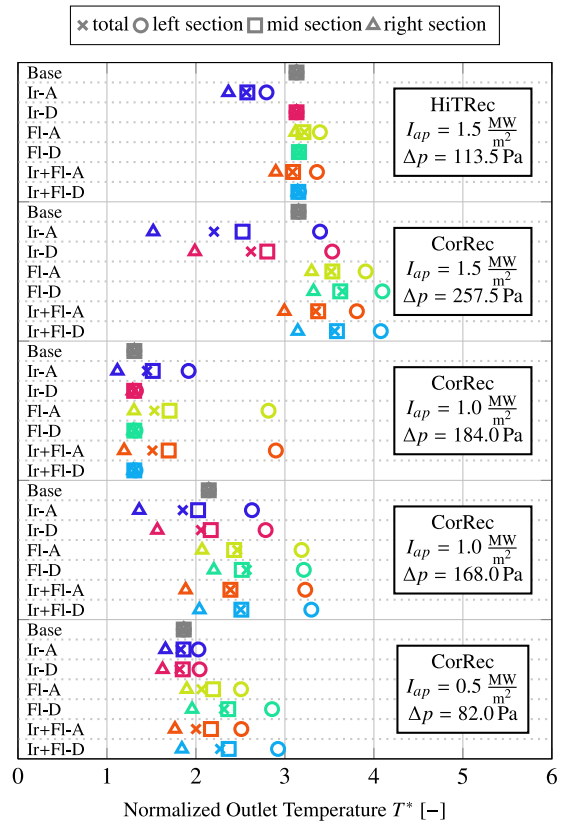


Fig. 2. Stage 2 results — normalized outlet temperatures for the 3D simulation cases for the HiTRec and CorRec absorbers.

pressure drop levels selected based on the non-monotonous regions in Fig. 1 instead show clear unstable flow behavior. Despite deactivated perturbations, i.e. despite uniform irradiation and uniform pressure boundary conditions, the outlet temperatures still vary significantly between the three sections.

The different flow behavior between the HiTRec and CorRec absorbers is further illustrated by Figs. 3 and 4. Each figure shows the solid surface temperature distributions for the base case, the case with an active combined perturbation (“Ir+Fl-A”), and the continued case with the deactivated perturbation (“Ir+Fl-D”). The HiTRec absorber successfully compensates the temperature increase in the left-most section caused by the applied disturbance, but a notable increase remains for the CorRec design. Concluding, the presented methodology

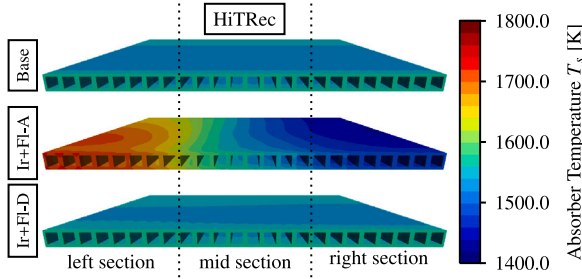


Fig. 3. Stage 2 results — 3D view of the solid surface temperature distributions for selected simulation cases for the HiTRec absorber ($I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$ & $\Delta p = 113.5 \text{ Pa}$).

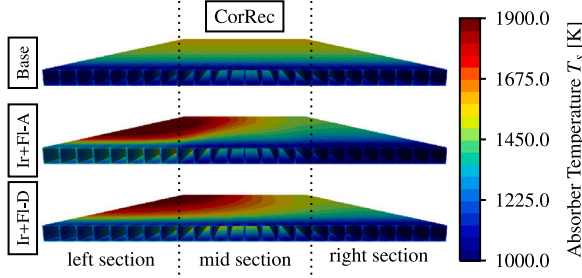


Fig. 4. Stage 2 results — 3D view of the solid surface temperature distributions for selected simulation cases for the CorRec absorber ($I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$ & $\Delta p = 257.5 \text{ Pa}$).

correctly evaluates the flow behavior as stable for the HiTRec absorber and unstable for the CorRec absorber.

3.2. Result independence

With the aim of demonstrating the robustness of the presented flow stability analysis, the key numeric parameters of the second analysis stage, the number of simulated absorber unit cells and the introduced irradiation and flow disturbances, are investigated as part of the next sets of simulations.

Number of simulated absorber unit cells

Through numerical simulations, Hoffschmidt (1997) showed that the tendency towards flow instability of a volumetric absorber module is related to the ratio of its length/depth L to its width W . The cross-sections relevant for lateral heat transfer are directly proportional to the absorber depth, while the distance over which disturbance have to be compensated is proportional to the absorber width. As a consequence, Hoffschmidt concluded that an absorber's flow stability generally increases³ with an increasing ratio $\frac{L}{W}$. All absorbers investigated within this publication have a fixed length of $L = 40 \text{ mm}$. However, the simulated absorber width directly depends on the number of simulated absorber unit cells. To test this dependency, the 3D simulations for the HiTRec and CorRec absorber for a flux level of 1.5 MW from the previous section were repeated for varying numbers of absorber unit cells. The resulting normalized outlet temperatures are shown in Fig. 5.

For the HiTRec absorber geometry, the base case and the cases with a combined irradiation and flow disturbance were carried out for one, three, eight and seventeen absorber channels per each of the three evaluation sections. The same disturbances, i.e. the same massflow

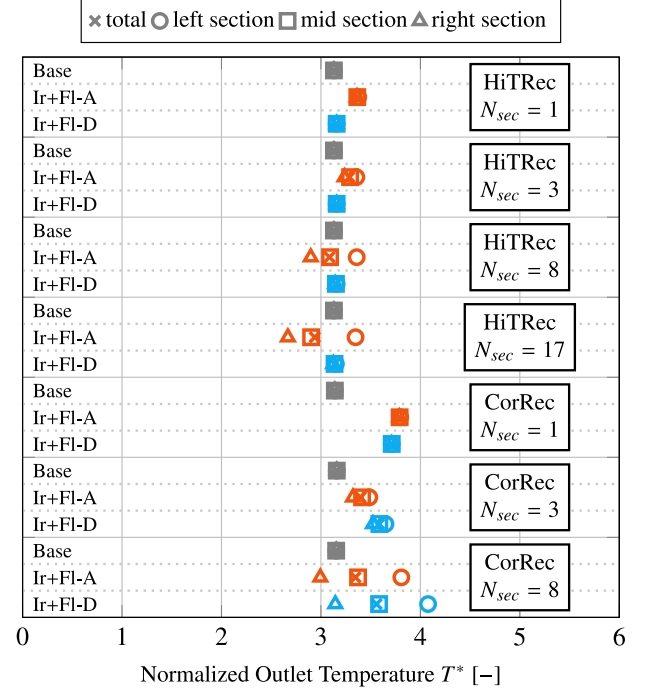


Fig. 5. Stage 2 results — normalized outlet temperatures for different numbers of unit cells for the HiTRec and CorRec absorbers ($I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$, $\Delta p = 113.5 \text{ Pa}$ for HiTRec, $\Delta p = 257.5 \text{ Pa}$ for CorRec).

reductions and irradiation changes, were applied for all simulations. The maximum of 51 total channels corresponds to a physical width 142.8 mm slightly larger than the width of 140 mm of a standard HiTRec absorber module, which is considered as the maximum relevant width. As expected based on the observations by Hoffschmidt (1997), the impact of the active disturbances (cases “Ir+Fl-A”) increases with a growing number of absorber unit cells. However, the HiTRec absorber is able to negate the effect of the perturbations successfully after their deactivation (cases “Ir+Fl-D”) clearly indicating a stable flow behavior. The simulations of the CorRec absorber were repeated for one, three and eight channels per section with a maximum simulated physical width of 30 mm .

Compared to HiTRec, a significantly greater influence of the disturbances can be observed despite the smaller corresponding physical width. In the case of $N_{sec} = 1$, the three simulated absorber channels show homogeneous outlet temperatures, but the outlet temperature level is changed even after the deactivation of the perturbation compared to the base cases. This illustrates that the CorRec design is able to homogenize perturbations over the small width of three channels, but the perturbation can change the operating point of the whole absorber due to its unstable flow behavior. For $N_{sec} = 8$, the CorRec absorber cannot compensate the disturbance, the evaluations section maintain their non-uniform outlet temperatures. Even larger configurations do not need to be investigated as the effect of the flow instability only increases with the simulated width.

The general approach of the flow stability analysis should include the simulation of an increasing number of absorber channels until either a clear indicator for flow instabilities is found (e.g. CorRec) or stable behavior is observed even for the largest relevant physical width (e.g. HiTRec).

Irradiation and flow disturbances

The irradiation and flow disturbances are meant to test the volumetric absorbers against non-homogeneous conditions caused by e.g. cloud-passages or a temporary blockage of absorber channels. The numerical

³ The pressure drop characteristic of an absorber is also affected by a change of the absorber depth, which could potentially counteract the increased flow stability.

Table 3

Overview of the tested disturbance configurations.

Set	Initial values		Irradiation disturbance	Disturbed section (flow)
	T_f [K]	T_s [K]		
1	800.00	1000.00	$\pm 10\%$	Left
2	373.15	473.15	$\pm 3\%$	Left + mid
3	373.15	1200.00	$\pm 5\%$	Left + right

implementation of the perturbations however offers a number of degrees of freedom, e.g. the strength of the irradiation change or the scope of the massflow reduction. Three sets with different perturbations were therefore tested to analyze the influence on the flow stability analysis. The first set of simulations corresponds to the simulations shown in Section 3.1 ($I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$). For the other two sets the form of the irradiation disturbance (cases “Ir-A”) was kept the same, i.e. an increased flux level for the left section, an unchanged level for the mid section, and a decrease for the right section. However, the strength of the irradiation change was reduced from $\pm 10\%$ for set 1 to $\pm 3\%$ for set 2 and $\pm 5\%$ for set 3. As described in Section 2.2, the flow disturbance for the first simulation set consists of a significant reduction of the massflow through the left-most section while keeping the base case massflow level for the other two sections. For the second simulation series, the massflow reduction is applied to the left and mid section and only the right section is kept at the base level. The third flow disturbance is created through a massflow reduction for the left section, a massflow increase for the right section and the base level for the mid section. The uniform initial temperature distributions used for the base cases and the cases with active perturbations are also modified between the three sets as shown in Table 3.

In Fig. 6, the normalized outlet temperatures are presented for the different perturbation sets for both absorber geometries. The HiTRec absorber consistently compensates all disturbances and always converges towards the base operating point after the deactivation of the disturbances. The temperature distributions for the CorRec absorber differ for the three simulation sets, but all show the apparent unstable flow behavior of the CorRec design. The goal of the presented flow stability analysis is to determine if an absorber geometry exhibits the tendency towards flow instability or not rather than simulating a specific unstable scenario. All tested perturbation configurations are equally suitable to test an absorber’s stability as seen in Fig. 6.

3.3. 3D-shaped absorbers

The presented methodology was developed with the aim of assessing the flow stability of 3D-shaped volumetric absorbers. 3D-Shaped absorbers are characterized by intentional changes in their geometry along the flow direction. In particular, 3D-shaped honeycomb absorbers often have openings in the channel walls, which allow lateral energy exchange via radiation or interchanging air flow on top of the lateral heat conduction within the solid absorber. To investigate the influence of such geometric features on the flow stability, two 3D-shaped adaptations of the CorRec absorber were created. A single absorber channel for each is shown in Fig. 7. The first design, the CorStep absorber, is based on the StepRec absorber investigated in a previous publication (Broeske et al., 2022).

The StepRec absorber contains three steps of increasing width at the absorber front, which allow for a deeper penetration of the incident solar radiation and improve the thermal efficiency. The CorStep absorber is given a single step with a length of 5 mm and a width equal to half the channel width. The idea of the short step is to keep the pressure loss characteristic of the CorStep absorber close to the one of the original CorRec geometry, while still enabling the investigation of the effect of the open channel walls on the flow stability. The step is only added in the horizontal direction, so that the 3D stage of the analysis can again be limited to a horizontal row of absorber channels

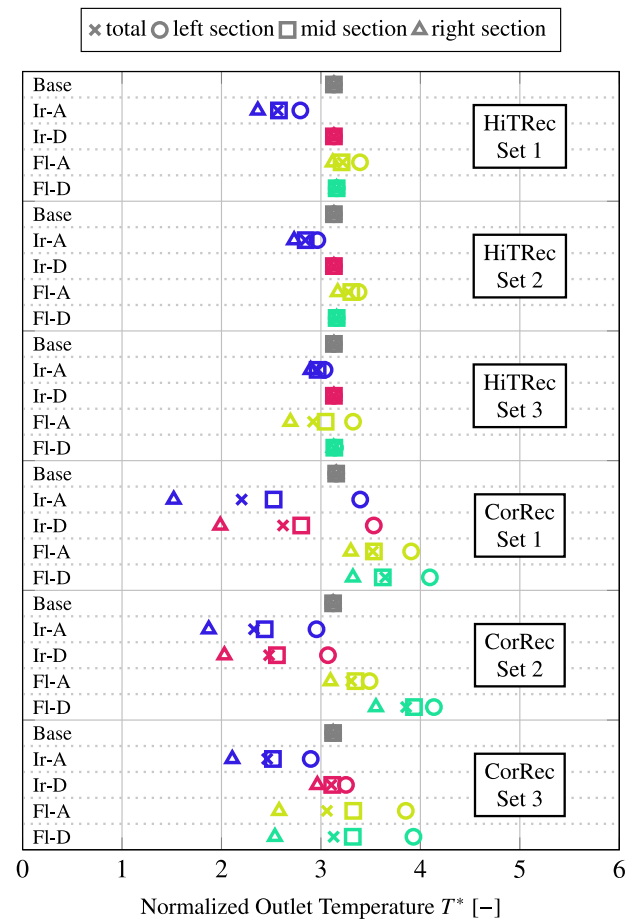


Fig. 6. Stage 2 results — normalized outlet temperatures for different configurations of irradiation/flow disturbances for the HiTRec and CorRec absorbers ($I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$, $\Delta p = 113.5 \text{ Pa}$ for HiTRec, $\Delta p = 257.5 \text{ Pa}$ for CorRec).

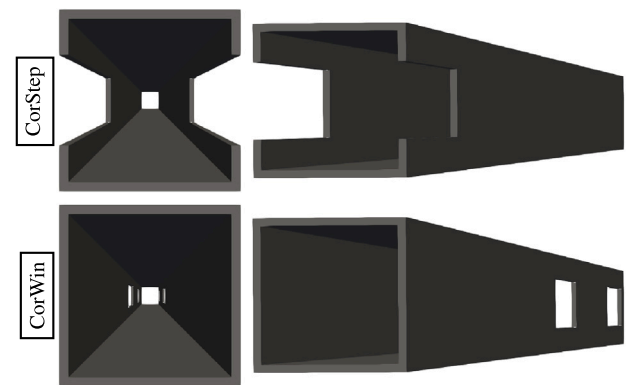


Fig. 7. Unit cell models for the CorStep and CorWin absorbers.

instead of a full square. The second adaption, CorWin, adds four alternating windows to the downstream half of the absorber channel. As seen in Fig. 4, the effect of the perturbations, which the CorRec absorber cannot compensate for, is notably focused in the deeper half of the absorber channels. The four windows are added in an alternating pattern into the right and left side walls of the absorber channels. This makes a direct radiative energy exchange between channels and their second next neighbors possible and potentially reduces the effect of the perturbations. The wall strength, channel width and heat conductivity are kept unchanged from the original CorRec design (cf. Section 3.1).

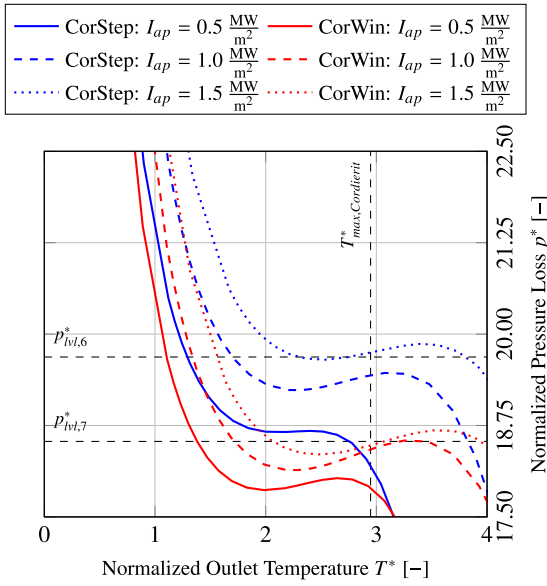


Fig. 8. Stage 1 results — normalized pressure loss over normalized outlet temperatures for the CorStep and CorWin absorbers.

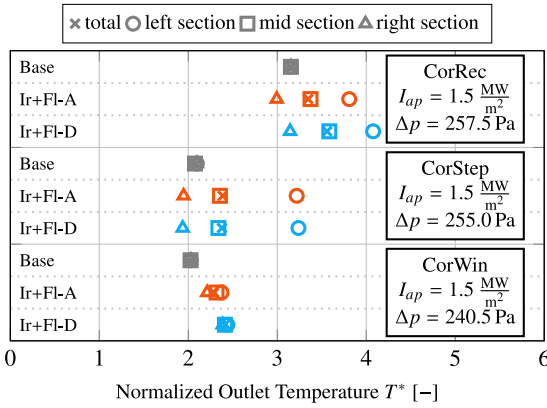


Fig. 9. Stage 2 results — normalized outlet temperatures for the 3D simulation cases for the CorStep and CorWin absorbers compared to the CorRec results.

The results of the continuum model simulations for the CorStep and CorWin absorbers are given in Fig. 8. The same incident solar radiation cases as in Section 3.1 were considered. Similar to the CorRec results (cf. Fig. 1), both designs show zones of non-monotonous pressure loss behavior signaling potential for flow instability. Therefore, one pressure drop level per geometry and $I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$ were selected for the 3D stage of the flow stability analysis. As for lateral heat conduction, the effect of energy exchange across channel openings on flow stability can only be assessed via the second 3D stage, because the continuum model assumes laterally uniform temperature distributions. For the CorStep absorber, a new pressure drop level $p_{lvl,6}^* = 19.6$ ($\Delta p_{lvl,6} = 255.0 \text{ Pa}$) close to the level of the CorRec absorber ($p_{lvl,3}^*$) is used due to the similar pressure loss characteristics. A seventh pressure loss level $p_{lvl,7}^* = 18.5$ ($\Delta p_{lvl,7} = 240.5 \text{ Pa}$) has been selected for the CorWin geometry.

In Fig. 9, the normalized outlet temperatures for the different channel sections are presented for the base cases and the cases of a combined irradiation and flow perturbation. Perturbations similar to the simulations of CorRec absorber were applied, the corresponding outlet temperatures for the CorRec design are repeated in Fig. 9. The solid surface temperature distributions for the CorStep absorber are

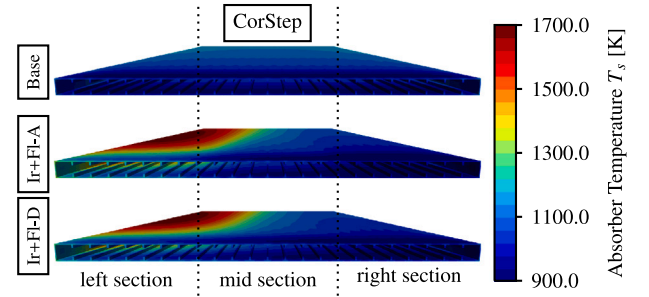


Fig. 10. Stage 2 results — 3D view of the solid absorber temperature distribution for selected simulation cases for the CorStep absorber ($I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$ & $\Delta p = 255.0 \text{ Pa}$).

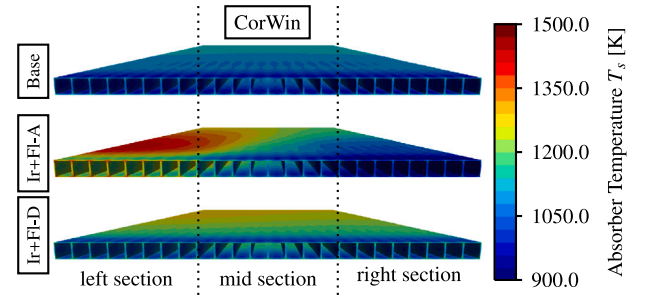


Fig. 11. Stage 2 results — 3D view of the solid surface temperature distributions for selected simulation cases for the CorWin absorber ($I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$ & $\Delta p = 240.5 \text{ Pa}$).

given by Fig. 10 and for the CorWin absorber by Fig. 11. Overall, the CorStep absorber shows the similarly unstable flow behavior compared to the original CorRec design. However, the CorWin absorber achieves homogeneous outlet temperatures even for the case with an active disturbance (“Ir+Fl-A”), as only the upstream half without the windows is affected. But, the unstable flow characteristic causes the operating point of all 24 channels to change for the case with the deactivated disturbance (“Ir+Fl-D”) despite the homogeneous outlet temperatures. The results for the two 3D-shaped absorbers lead to the expectation that further modification, e.g. a combination of the two designs, could yield a stable volumetric absorbers even with the low heat conductivity of Cordierite.

4. Discussion

The presented flow stability analysis methodology was shown to be able to correctly identify the flow behavior of the HiTRec absorber as stable and the CorRec absorber’s behavior as unstable. The numeric simulation results therefore match the observations made during experiments in the past by Hoffschmidt et al. (2001) and Fend et al. (2004). This also has to serve as the validation of the new approach due to the lack of further, appropriate experimental data.

As seen by the two pressure drop levels which were investigated for the CorRec design at $I_{ap} = 1.0 \frac{\text{MW}}{\text{m}^2}$, the flow instabilities only occur within certain, critical pressure loss regions ($\Delta p_{lvl,2} = 168.0 \text{ Pa}$). Outside of these zones ($\Delta p_{lvl,4} = 184.0 \text{ Pa}$), the flow behavior of the CorRec absorber appears as stable. Thus, the first step of the flow analysis is necessary to identify these critical pressure loss levels. In turn, the second stage using the 3D CFD model is required to correctly assess the influence of lateral energy exchange on flow stability. The general pressure loss characteristic for the CorRec (Fig. 1) and the CorStep (Fig. 8) absorbers exhibit similar critical, non-monotonous regions. But, the open channel walls of the CorStep design make it better suited to compensate non-homogeneous temperature distribution associated

with flow instability (cf. Fig. 9), which can only be observed via the 3D CFD model.

In accordance with the results of Hoffschmidt (1997), the likelihood of flow instability was seen to increase for an increasing width of the investigated absorber segment (cf. Fig. 5). The considered absorber width is directly proportional to the number of simulated absorber unit cells during the second analysis stage. Therefore, a robust analysis should include simulations of an increasing number of absorber channels until either a clear indicator for flow instabilities is found or stable behavior is observed even for the largest relevant physical width.

The implementation of the irradiation and flow stability disturbances offer a number of degrees of freedom to chose the actual disturbance against which the volumetric absorbers are tested against. All configurations considered for this publication appeared equally suited to evaluate the flow stability (cf. Fig. 6). However, it is difficult to directly assess how likely a concrete perturbation is to occur in a real open volumetric receiver. In the numerical analysis, the HiTRec absorber, which is known to be stable based on experiments and the operation of the Solar Tower Jülich, is able to compensate even the stronger perturbations, e.g. an irradiation disturbance of $\pm 10\%$ at $I_{ap} = 1.5 \frac{\text{MW}}{\text{m}^2}$ across a few absorber channels. Therefore, it is deemed advantageous to intentionally select strong disturbances in order to prevent an underestimation of flow instabilities, while using the HiTRec absorber as the benchmark for stable volumetric absorbers.

5. Conclusion and outlook

A new, numerical methodology for the flow stability analysis of volumetric absorbers has been developed. The first stage of the methodology applies an 1D LTNE continuum model to quantify the general pressure loss characteristic of an absorber and identifies critical pressure drop levels with potential for flow instability. These levels are tested as part of the second analysis stage using a discrete 3D CFD model against different irradiation and flow perturbations. If the absorber is able to compensate the disturbances even for the critical pressure drop levels, its flow behavior is deemed stable. Otherwise, the absorber geometry is considered susceptible to flow instability.

The new methodology has been validated for two reference absorbers with known stable resp. unstable flow behavior. Further, the new approach was applied to analyze the flow stability of two 3D-shaped honeycomb absorber geometries. It was shown that the geometric features of such 3D-shaped absorbers can directly improve the flow stability, but do not prevent instability. Therefore, a flow stability analysis should be included in design/optimization processes of new volumetric absorbers alongside a maximization of the thermal efficiency. The new simulation approach was specifically created for this purpose. However, this cannot not directly replace the experimental evaluation of the flow stability, more experimental data would be particularly useful for 3D-shaped absorber designs. As the next steps, the presented methodology is planned to be further tested for and to be applied to more complex 3D-shaped volumetric absorber geometries.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The research institution at which this work has been conducted (DLR) holds a number of patents in relation to the topic of this publication (volumetric air receivers, “HiTRec” vol. air receiver technology) including: DE10029940C1, DE19740644C2, DE112017002386A5. Some of the patents have been licensed (non-exclusively) to the company Kraftanlagen München GmbH.

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Appendix. 3D CFD model details

While Section 2.2 focused on the specific modeling of the flow instabilities, the different components of the 3D CFD model are explained in this Appendix. The model is implemented within the open-source CFD software OpenFOAM® (2022). The majority of the model was already available in OpenFOAM, only the main boundary condition (Eq. (A.4)) had to be added. The raytracing software which is used for the simulation of the incident solar irradiation is not part of OpenFOAM, but was developed at the German Aerospace Center in the past.

Governing equations

The air flow is modeled based on the stationary, laminar and compressible Navier–Stokes equations, which in general vector notation are given by (based on Schröder, 2010):

$$\begin{cases} \nabla \cdot (\rho \mathbf{u}) &= 0 \\ \nabla \cdot (\rho \mathbf{u} \mathbf{u}^T) &= -\nabla p + \nabla \cdot \boldsymbol{\sigma} \\ \nabla \cdot (\rho \mathbf{u} h) + \nabla \cdot (\rho \mathbf{u} K) &= \nabla \cdot (\boldsymbol{\sigma} \cdot \mathbf{u}) + \nabla \cdot (\kappa_f \nabla T_f) \end{cases} \quad (\text{A.1})$$

Here, $\mathbf{u}$ is the velocity vector, $K = \frac{1}{2} \mathbf{u} \cdot \mathbf{u}$ denotes the (specific) kinetic energy and $\boldsymbol{\sigma}$ the linear stress tensor for Newtonian fluids:

$$\boldsymbol{\sigma} = -\frac{2}{3} \mu \nabla \cdot \mathbf{u} \mathbf{I} + \mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \quad (\text{A.2})$$

The Navier–Stokes equations are solved for the void volume of the horizontal row of absorber unit cells plus an (empty) entry region in front of the channels. The air density is modeled using the ideal gas law, while temperature and pressure dependent functions are implemented based on data from Stephan et al. (2013) for the other air properties.

For the solid absorber volume of the unit cell row, a general heat conduction equation (e.g. Baehr and Stephan, 2006) is solved for the solid temperature field T_s :

$$0 = \nabla \cdot (\kappa_s \nabla T_s) \quad (\text{A.3})$$

Temperature-dependent experimental data from the German Aerospace Center is used for the heat conductivity of SiSiC, while the heat conductivity of Cordierit is set based on data from Hoffschmidt (1997). The differential equations are solved using the finite volume method as implemented by OpenFOAM. Second-order, upwind-biased solution schemes are used for the air flow simulation and second order central differencing schemes for the solid heat conduction.

Boundary conditions

The different components of the 3D model are connected with each other through an energy balance for the absorber surface areas:

$$0 = \dot{q}_{cond,abs} + \dot{q}_{cond,air} + \dot{q}_{sol} - \dot{q}_r \quad (\text{A.4})$$

The energy balance is used to calculate the distribution of the surface temperature T_{surf} , which is then applied as a fixed value boundary condition for both volumetric regions, i.e. for the simulation of the air flow and the simulation of the heat conduction within the absorber. The derivation of the absorbed solar flux $\dot{q}_{sol}$ and the net radiative flux $\dot{q}_r$ are explained in the next section. The heat conduction between solid absorber and absorber surface ($\dot{q}_{cond,abs}$) and between air flow and absorber surface ($\dot{q}_{cond,air}$) are evaluated based on the surface normal temperature gradients. These gradients depend on the surface temperature T_{surf} and the volumetric temperature distributions of the solid (T_s) resp. the air (T_f) temperature. As a result, Eq. (A.4) connects all model components which are solved in an iterative fashion. Only the absorbed solar flux is calculated beforehand and treated as constant while solving the main model.

For the beginning of the entry region, a fixed value temperature boundary condition sets the inlet temperature T_{in} (cf. Eq. (1)), while zero gradient temperature boundary conditions are prescribed for all

remaining boundaries of the air flow and the heat conduction simulation, i.e. for the absorber outlet, the entry region sides and the outer sides of the solid absorber volume.

The flow boundary conditions differ between the simulation cases with active flow disturbances (cases “Fl-A” and “Ir+Fl-A”, cf. Table 1) and all other 3D simulation cases. For the cases without flow disturbances, pure pressure boundary conditions are used. At the start of the entry region, the ambient/inlet pressure value of p_{in} is applied, while the pressure at the outlet of each of the N_{tot} absorber channels is set to $p_{out} = p_{in} - \Delta p$ depending on the selected pressure drop level. A zero gradient boundary condition is used for the velocity at the absorber outlet, while the inlet velocities are derived from the pressure gradient. For the cases with active flow disturbances, the outlet pressure boundary condition is changed to a zero gradient boundary condition. Outflow velocities based on the massflow density for the three channel sections are prescribed causing the flow disturbances. The deactivation of the flow disturbances (cases “Fl-D” and “Ir+Fl-D”) is equal to reverting the flow boundary conditions back to the purely pressure-driven boundary condition set. The pressure-driven boundary set in particular allows the 3D model to calculate the resulting massflow density distributions for the N_{tot} absorber channels, which can range from uniform in case of a stable absorber (cf. Fig. 3) to non-uniform in case of an unstable absorber (cf. Fig. 4).

Radiation modeling

All absorber surfaces are treated as opaque, diffusive reflecting surfaces with constant optical properties. For the SiSiC material of the HiTREC absorber and for the Cordierite material of the CorRec, CorStep and CorWin designs, a surface absorptivity and emissivity of $\alpha_{surf} = \epsilon_{surf} = 80\%$ is used based on Hoffschmidt (1997). The radiation modeling is split into separate approaches for the incident solar irradiation and the thermal radiation exchange between the absorber surfaces.

The raytracing software FEMRay is used to track the incident solar irradiation. FEMRay was developed at the German Aerospace Center as an extension of the raytracing software MIRVAL and allows surface definitions via finite element meshes with MIRVAL (cf. Amsbeck et al., 2008; Schwarzbözl et al., 2008). The starting positions of the solar rays are spread out (based on a uniform random distribution) across the void portions of the channel entry cross-sections, i.e. the void part of cross-section containing the absorber front surface. The initial directions are also generated randomly based on an average initial direction and an opening angle. The opening angle defines a cone around the average direction from which the initial directions are sampled as deviations from the average direction. The absorption and diffusive reflection of solar rays throughout the porous absorber are then tracked until the rays are either totally absorbed or reflected (back) out of the absorber structure. The final result of the raytracing software is the 3D distribution of the absorbed solar irradiation for each absorber surface $\dot{q}_{sol}$, which is used as part of the boundary Eq. (A.4).

The thermal emissions of the absorber surfaces and the subsequent radiative heat exchange is modeled via a surface-to-surface radiation model based on viewfactors. The radiation model, which is already part of OpenFOAM, defines the net radiative flux of each surface $\dot{q}_r$ as the difference between irradiance G and surface radiosity J . Based on the numeric division of the absorber surfaces into discrete subsurfaces or mesh elements, the surface radiosity of one element k is the sum of surface emission and reflected irradiance, while the irradiance is calculated from the surface radiosities of all other elements and the viewfactors $\Psi_{k,l}$ between the elements:

$$\dot{q}_{r,k} = G_k - J_k, \quad (A.5)$$

$$J_k = \epsilon_k \sigma_{sb} T_k^4 + (1 - \epsilon_k) G_k, \quad (A.6)$$

$$G_k = \sum_l \Psi_{k,l} J_l \quad (A.7)$$

In addition to the absorber surfaces, the void portions of the channel entry cross-sections and the void absorber outlet cross-sections are also

part of the viewfactor model as artificial surfaces. The artificial inlet surfaces are treated as black surfaces ($\epsilon_k = 1$) with a temperature of zero Kelvin ($T_k = 0$). All radiation which reaches the inlet surfaces is absorbed and considered lost as part of the absorber (bulk) emission loss. The artificial outlet surfaces are also treated as black surfaces ($\epsilon_k = 1$), but they are given an artificial temperature $T_{out,k}$ which is updated throughout the simulation:

$$T_{out,k} = \sqrt[4]{\frac{G_{out,k}}{\sigma_{sb}}} \quad (A.8)$$

This artificial temperature results in a net radiative flux of zero for the outlet surfaces implementing a perfectly (diffusive) reflective boundary condition for the absorber outlet. (Setting an emissivity of $\epsilon_k = 0$ is not possible as the derivation of the solution via a linear system of equations requires $\epsilon_k > 0$.) The radiation elements corresponding to the absorber surfaces are given the material emissivity ($\epsilon_k = \epsilon_{surf}$) and their temperature is set based on the surface temperature distribution calculated based on the boundary Eq. (A.4).

A system of linear equations can be derived from Eqs. (A.5)–(A.7) which is solved for the net radiative flux. The net radiative flux is then used as part of the absorber surface boundary condition linking the radiation model with the other model components.

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